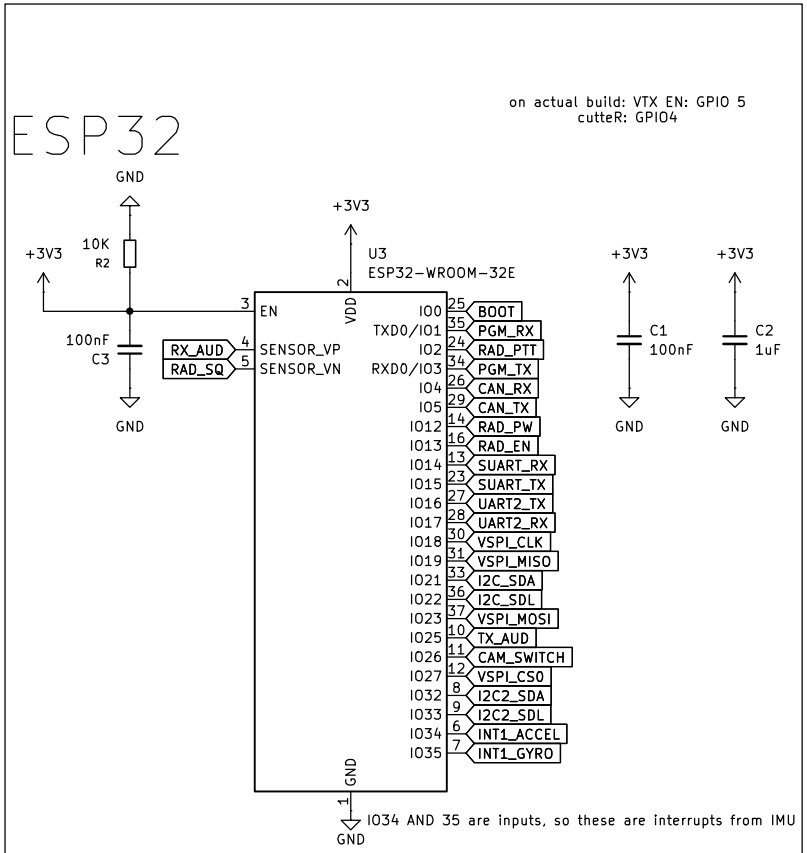
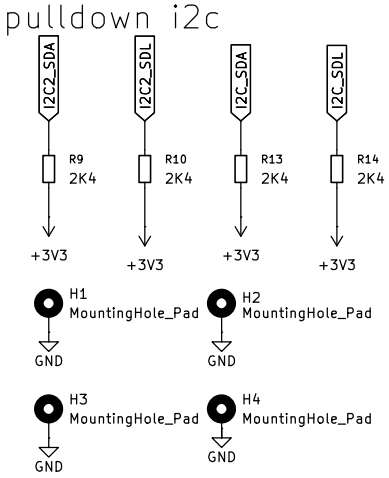
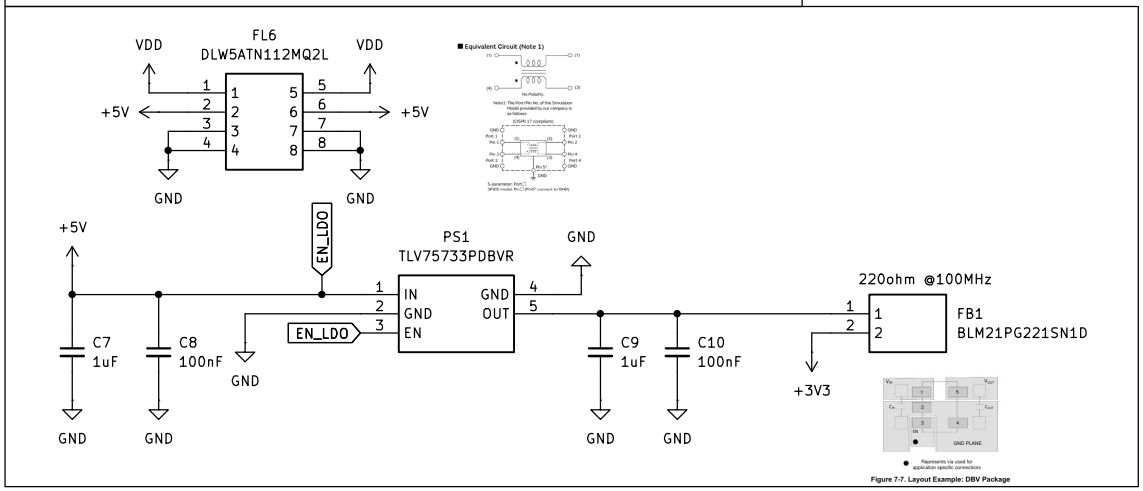






SENSORS

File: COMMS_BOARD_SENSORS.kicad_sch

Radio

File: COMMS_BOARD_RADIO.kicad_sch

CAN BUS

File: COMMS_BOARD_CAN_BUS.kicad_sch

FILTERING AND EMI MITIGATION

File: COMMS_BOARD_EMIFIL.kicad_sch

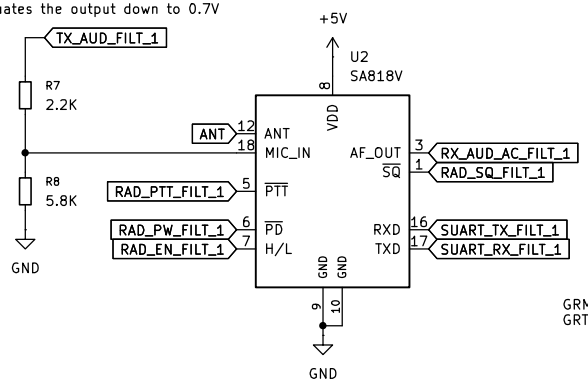
EXTERNAL CONNECTIONS

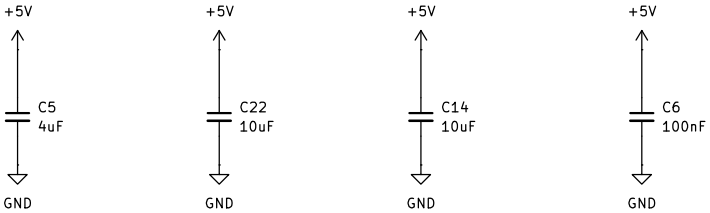
File: COMMS_BOARD_EXTERNAL_CONNS.kicad_sch

SA818S

Rad_EN is logic low enable to change to low power mode
Leave floating or open for high power mode
Rad_PW is logic low for power down and high for power up
Rad_PTT "0" force the module to enter TX state; and "1" to Rx state
*Before enter sleep mode, user need to pull low RXD Pin
to prevent current leakage or poor reset in the next time.

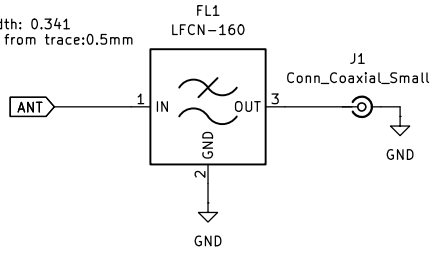
this attenuates the output down to 0.7V



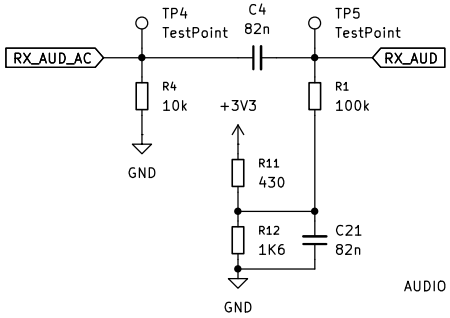


GRM219R61A475KE32D 0805 for 10% 4.uF @16VDC x5r
GRT21BR61E106KE13L 0805 for 10% 10uF @25VDC x5r

Trace Width: 0.341
No pour distance from trace:0.5mm



Rx audio bias+

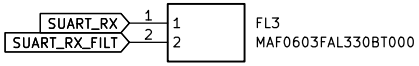


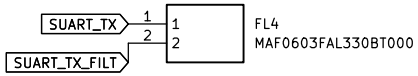
AUDIO OUTPUT IMPEDENCE: 200 OHM, OUTPUT AMPLITUTDE: 700mV

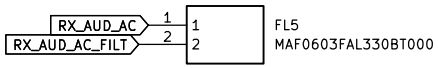
NOTE: ONLY DATA LINES ARE FILTERED
Not required for GPIO lines like:
1.RAD_PTT
2.RAD_PW
3.RAD_SQ

Signal Line Ferrite Bead

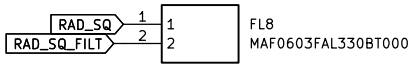






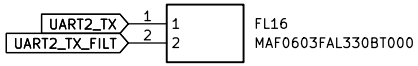


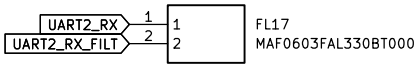




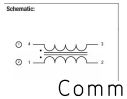


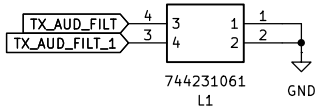


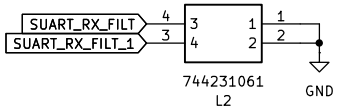


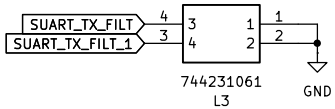


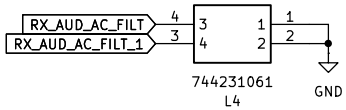
Common Mode Choke

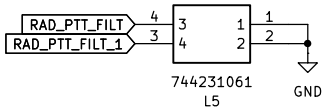


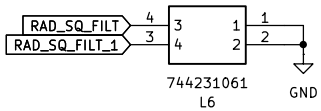


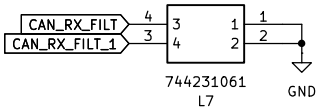


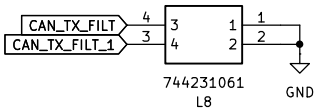


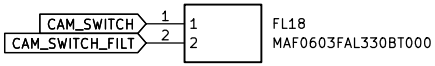


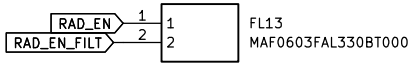




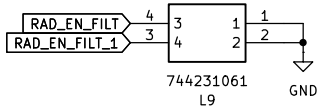


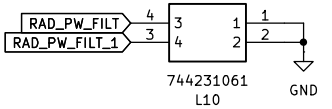




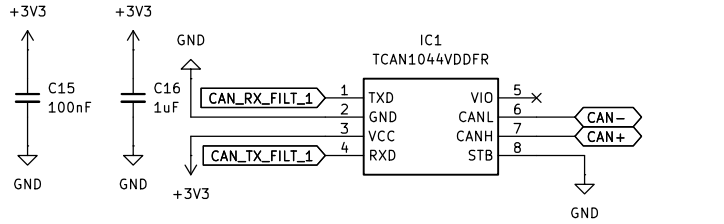






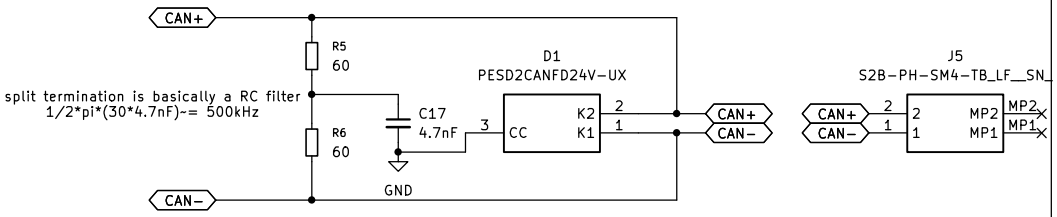


CAN TRANSCEIVER

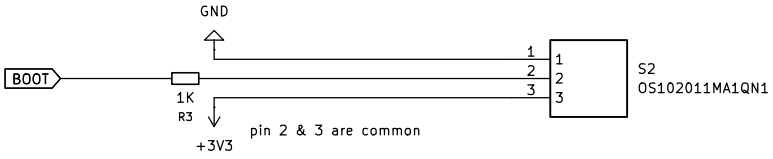


TO OPERATE IN NORMAL MODE, PULL STB TO GND

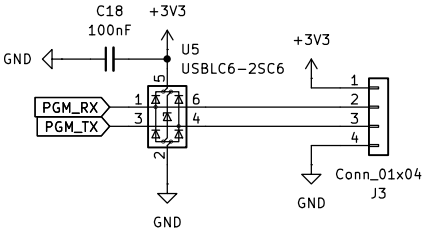
esd,emi filtering



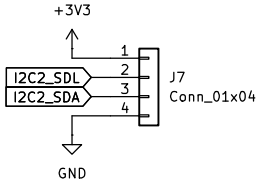
BOOT MODE SWITCH



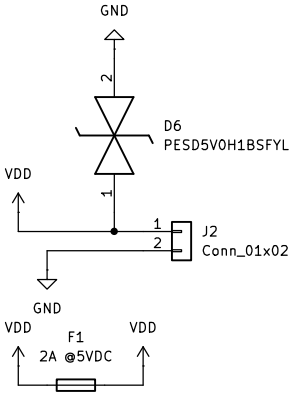
PROGRAMMING



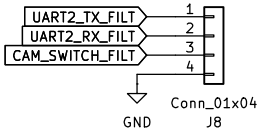
current sense+ power control

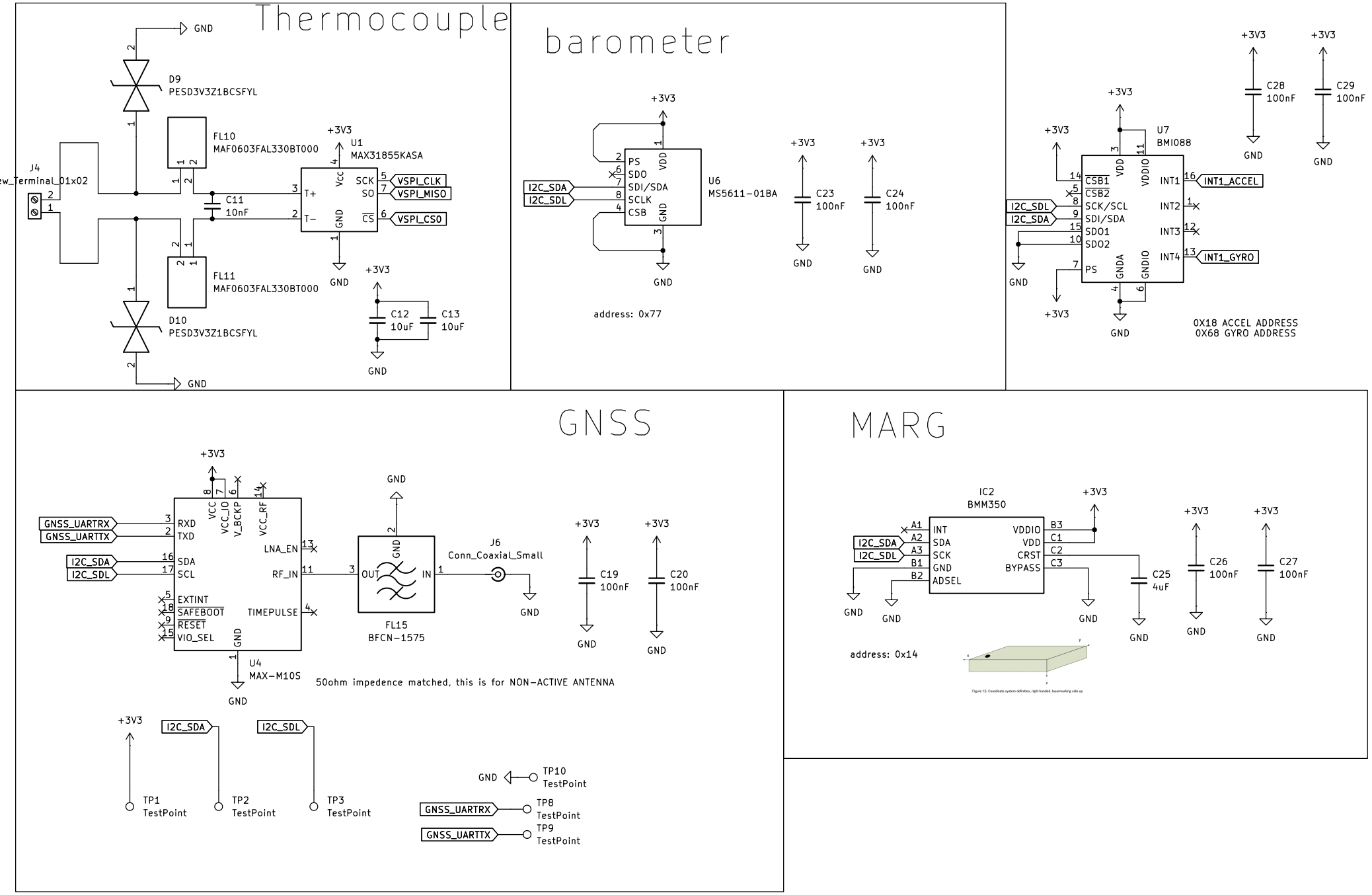


3V6 BUS POWER CONNECTOR



OSD+CAM SWITCHER




Thermocouple
GND
D9
PESD3V3Z1BCSFYL
FL10
MAF0603FAL330BT000
J4
Screw_Terminal_01x02
C11
10nF
+3V3
U1
MAX31855KASA
T+
T−
Vcc
GND
SCK
SO
CS
VSPI_CLK
VSPI_MISO
VSPI_CS0
FL11
MAF0603FAL330BT000
D10
PESD3V3Z1BCSFYL
C12
10uF
C13
10uF
barometer
U6
MS5611−01BA
PS
SDO
SDI/SDA
SCLK
CSB
VDD
I2C_SDA
I2C_SDL
C23
100nF
C24
100nF
address: 0x77
U7
BMI088
VDDIO
CSB1
CSB2
SCK/SCL
SDI/SDA
SDO1
SDO2
GNDA
GNDIO
INT1
INT2
INT3
INT4
INT1_ACCEL
INT1_GYRO
C28
100nF
C29
100nF
0X18 ACCEL ADDRESS
0X68 GYRO ADDRESS
GNSS
U4
MAX−M10S
VCC
VCC_IO
V_BCKP
VCC_RF
RXD
TXD
SDA
SCL
EXTINT
SAFEBOOT
RESET
VIO_SEL
LNA_EN
RF_IN
TIMEPULSE
GNSS_UARTRX
GNSS_UARTTX
FL15
BFCN−1575
OUT
IN
J6
Conn_Coaxial_Small
C19
100nF
C20
100nF
50ohm impedence matched, this is for NON−ACTIVE ANTENNA
TP1
TestPoint
TP2
TestPoint
TP3
TestPoint
TP10
TestPoint
TP8
TestPoint
TP9
TestPoint
MARG
IC2
BMM350
INT
SDA
SCK
GND
ADSEL
VDDIO
VDD
CRST
BYPASS
C25
4uF
C26
100nF
C27
100nF
address: 0x14